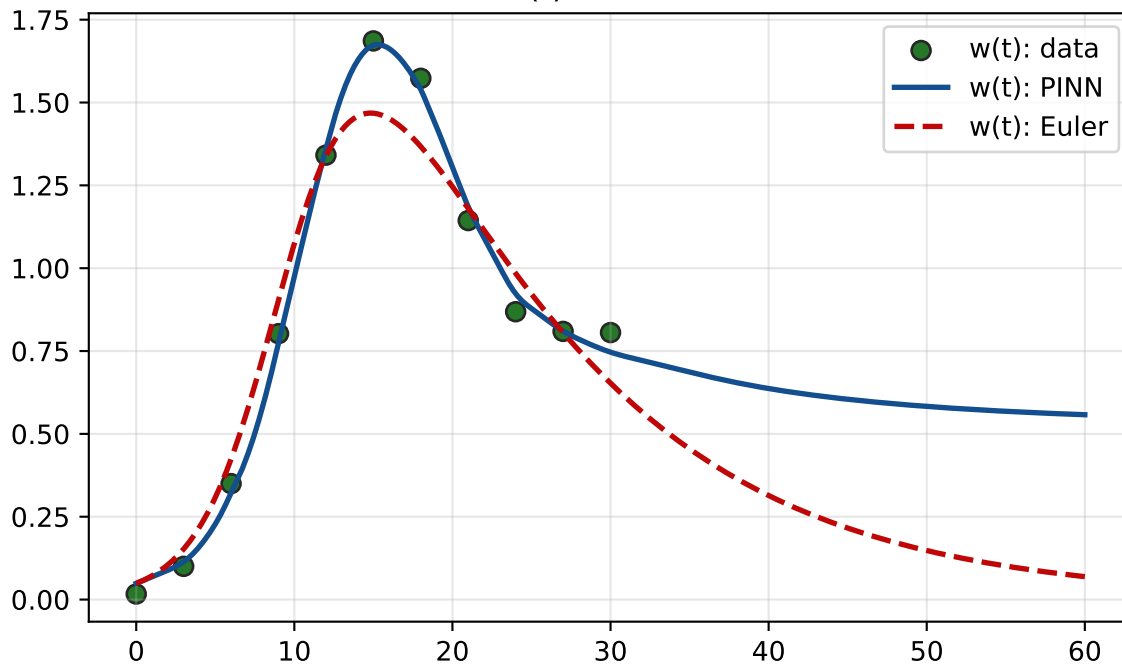
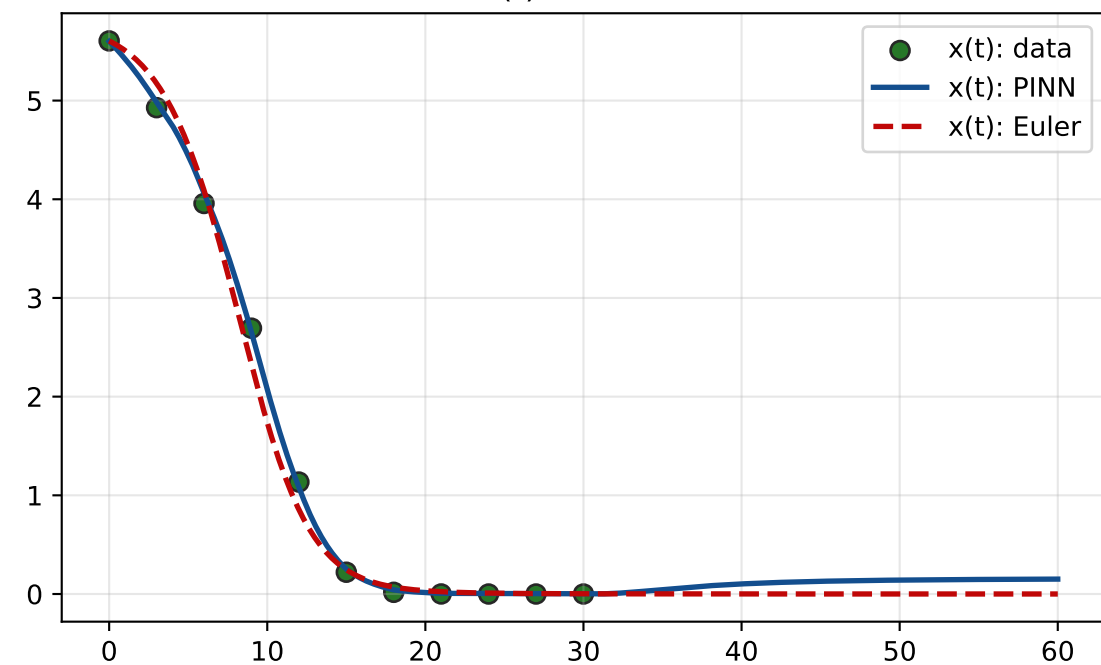


## E6 | PINN vs Euler

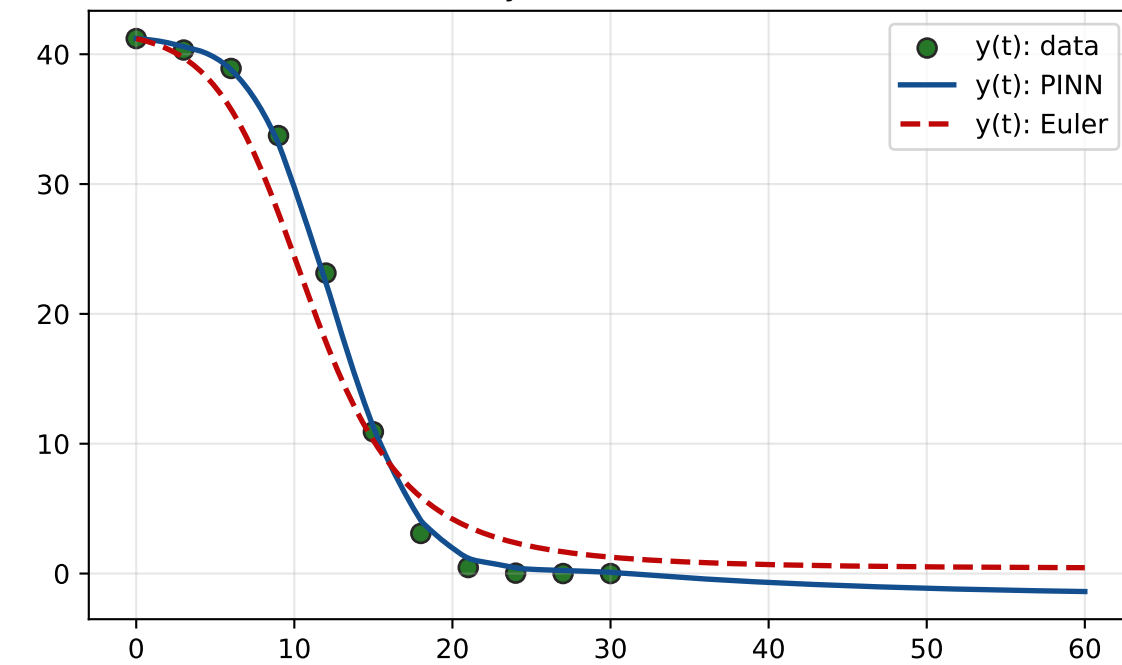
w(t): Biomass



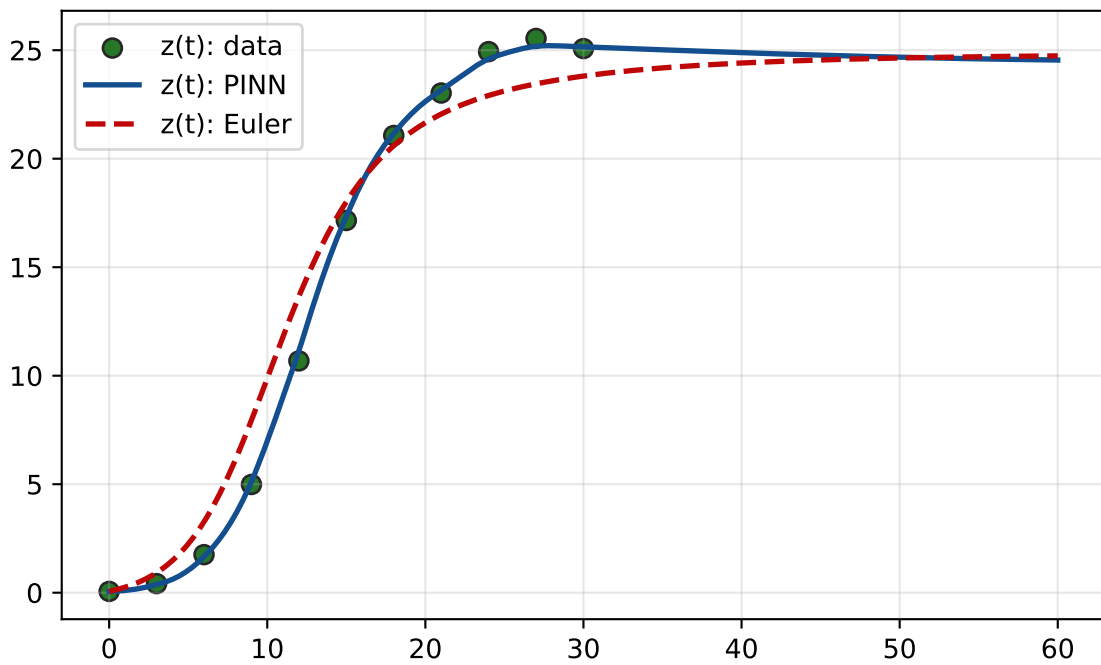
x(t): Glucose



y(t): Fructose



z(t): Ethanol



E6

R2=0.9803, R2\_adj=0.9765, RSS=1.348e+02, AIC=6.638e+01

cond(JTJ)=8.58e+05, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 8.407306e-03 | 2.045728e-03 | 4.145039e-03 | 4.262267e-03  | 1.255234e-02 | 2.108492e-04 |
| p2    | 4.611247e-02 | 1.487811e-01 | 3.014591e-01 | -2.553466e-01 | 3.475715e-01 | 7.583493e-01 |
| p3    | 2.989935e-03 | 4.686763e-03 | 9.496284e-03 | -6.506350e-03 | 1.248622e-02 | 5.274331e-01 |
| p4    | 2.907595e-01 | 2.232875e-01 | 4.524235e-01 | -1.616640e-01 | 7.431830e-01 | 2.009047e-01 |
| p6    | 1.288261e-01 | 6.198320e-02 | 1.255899e-01 | 3.236175e-03  | 2.544160e-01 | 4.466294e-02 |
| p8    | 6.697439e-02 | 3.291918e-02 | 6.670059e-02 | 2.738016e-04  | 1.336750e-01 | 4.911416e-02 |
| p9    | 2.025887e-05 | 8.471051e-03 | 1.716398e-02 | -1.714372e-02 | 1.718424e-02 | 9.981047e-01 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 4.868766e-02 |
| x0 (fixed)     | 5.604531e+00 |
| y0 (fixed)     | 4.120490e+01 |
| z0 (fixed)     | 6.092072e-02 |
| W0 (fixed)     | 0.000000e+00 |